Python Activity-Based Learning
13. ATM Transaction Sequence

Simple student-style assignment | Easy English | Step-by-step solution

1. Problem Statement

Create a simple ATM program that processes a list of deposits and withdrawals. The program must update the balance and reject a withdrawal when there is not enough money.

2. Objectives

• Store the starting balance.

• Process more than one transaction using a loop.

• Update and display the final balance.

3. Assumptions

• The starting balance cannot be negative.

• A transaction is entered as D for deposit or W for withdrawal.

• Deposits and withdrawals must be positive amounts.

4. Algorithm (Step by Step)

1. Start.

2. Read the opening balance and number of transactions.

3. Repeat for each transaction.

4. Read transaction type and amount.

5. If it is a deposit, add the amount to the balance.

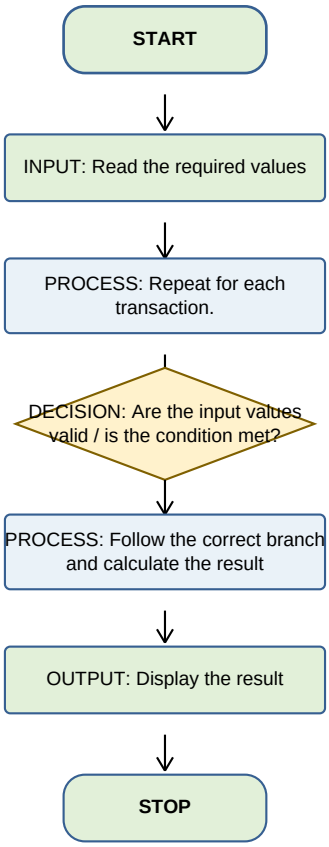
6. If it is a withdrawal, subtract it only when the balance is enough.

7. Otherwise, display a message that the transaction was rejected.

8. Display the final balance and stop.

5. Flowchart

Follow the arrows from START to STOP. The diamond shows a decision point.



6. Pseudocode

```
START
INPUT balance, n
REPEAT n TIMES
    INPUT type, amount
    IF amount <= 0 THEN PRINT Invalid amount
    ELSE IF type = D THEN balance = balance + amount
    ELSE IF type = W AND amount <= balance THEN balance = balance - amount
    ELSE IF type = W THEN PRINT Insufficient balance
    ELSE PRINT Invalid transaction type
PRINT balance
STOP
```

7. Python Source Code

```
balance = float(input("Enter opening balance: "))
n = int(input("How many transactions? "))

if balance < 0 or n < 0:
    print("Invalid input")
else:
    for i in range(n):
        kind = input("Enter D for deposit or W for withdrawal: ").upper()
        amount = float(input("Enter amount: "))

        if amount <= 0:
            print("Invalid amount")
        elif kind == "D":
            balance += amount
            print("Deposit successful")
        elif kind == "W":
            if amount <= balance:
                balance -= amount
                print("Withdrawal successful")
            else:
                print("Insufficient balance")
        else:
            print("Invalid transaction type")

    print("Final balance:", balance)
```

8. Test Cases

Input / Test	Expected result	What it checks
1000; D 500; W 200	1300	Deposit and withdrawal accepted
300; W 500	300	Withdrawal rejected
100; D 50	150	Deposit accepted

9. Sample Output

```
Enter opening balance: 1000
How many transactions? 2
Enter D for deposit or W for withdrawal: D
Enter amount: 500
Deposit successful
Enter D for deposit or W for withdrawal: W
Enter amount: 200
Withdrawal successful
Final balance: 1300.0
```

10. Conclusion

This program uses a loop and conditional statements to process ATM transactions and maintain the correct account balance.